

Changes in brain activation associated with reward processing in smokers and nonsmokers. A positron emission tomography study.

Tobacco smoking is the most frequent form of substance abuse. Several studies have shown that the addictive action of nicotine is mediated by the mesolimbic dopamine system. This system is implicated in reward processing. In order to better understand the relationship between nicotine addiction and reward in humans, we investigated differences between smokers and nonsmokers in the activation of brain regions involved in processing reward information. Using [$H_2(15O)$] positron emission tomography (PET), we measured regional cerebral blood flow (rCBF) in healthy smokers and nonsmokers while they performed a prelearned, pattern-recognition task. We compared two conditions involving nonmonetary reinforcement or monetary reward with a baseline condition in which nonsense feedback was presented. With monetary reward, we found activation in the frontal and orbitofrontal cortex, occipital cortex, cingulate gyrus, cerebellum, and midbrain in both groups. Additionally, monetary reward activated typical dopaminergic regions such as the striatum in nonsmokers but not in smokers. We found a similar pattern of activation associated with nonmonetary reinforcement in nonsmokers, whereas activation was found in smokers only in the cerebellum. The different patterns of activation suggest that the brains of smokers react in a different way to reward than those of nonsmokers. This difference involves in particular the regions of the dopaminergic system including the striatum. In principle these observations could be interpreted either as a consequence of tobacco use or as a primitive condition of the brain that led people to smoke. Supported by related nonimaging studies, we interpret these differences as a consequence of tobacco smoking, even if a short-term effect of smoking prior to the experiment cannot be excluded.